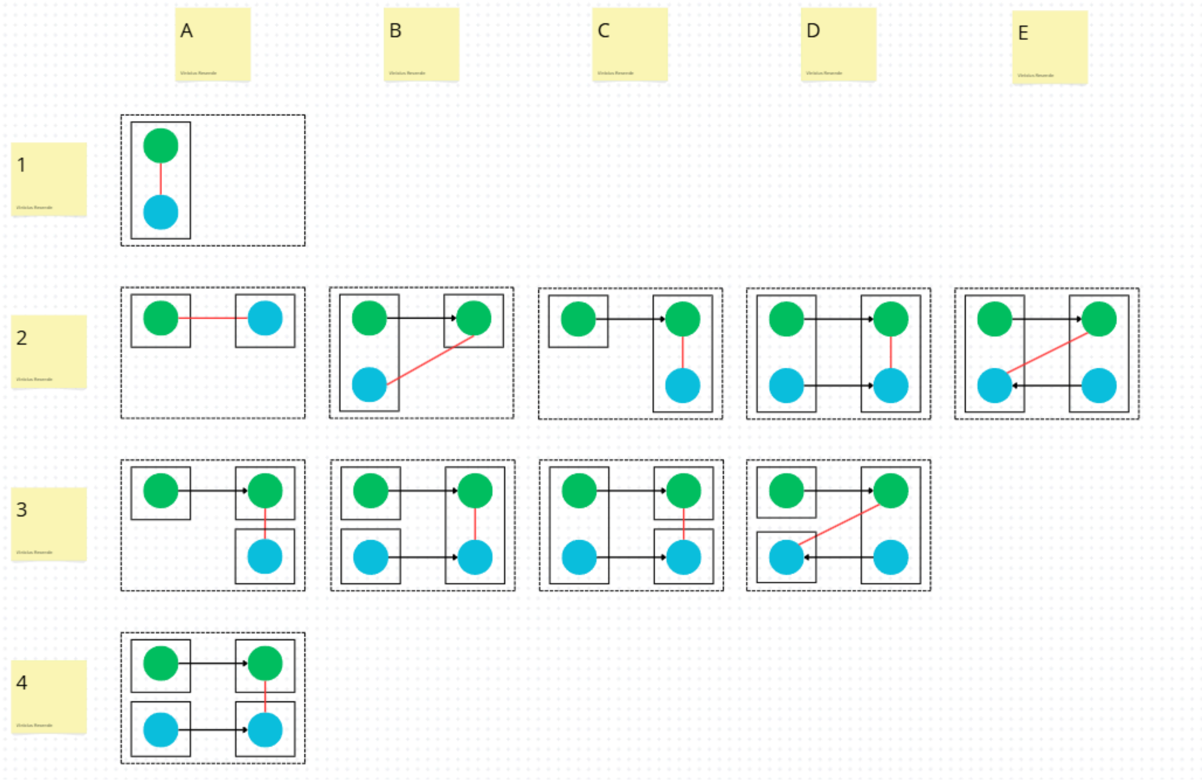
Override Assignment (OA)



A1 - Change of variable in left followed by change in right (in the same file)

src/main/java/samples/OverridingAssignmentLocalVariablesSample.java CHANGED

```
5   5        public static void conflict() {
    6  +         y = 2;    // left
6   7            x = x+1; // base
    8  +         y = 3;    // right
7   9            System.out.println(x);
8  10        }
9  11    }
```

On line 6, left changes the value of the variable 'y', then right does the same on line 8. This generates an OA interference from line 8 of 'OverridingAssignmentLocalVariablesSample.java' to line 6 of the same file.

A2 - Variable definition in left, modification in right (in different files)

Config.java

```java
1 // Arquivo: Config.java
2 public class Config {
3     public static int MAX_RETRIES;
4     static {
5         MAX_RETRIES = 5; // alteração de left
6     }
7 }
```

Main.java

```java
1 // Arquivo: Main.java
2 public class Main {
3     public static void main(String[] args) {
4         Config.MAX_RETRIES = 10; // Alteração de right
5         System.out.println("Max retries: " + Config.MAX_RETRIES);
6     }
7 }
```

In line 5 of the ‘Config.java’ file, left changed the value of the ‘MAX_RETRIES’ variable to 5. While in line 4 of the ‘Main.java’ file, right changed the value of the same variable to 10 before it was used.

B2 - Left adds a method call from file 1 to file 2. In file 2, a variable is defined/modified. In file 1, the Right method modifies this variable again.

ProcessadorDePedidos.java

```java
1  // Arquivo: ProcessadorDePedidos.java
2  public class ProcessadorDePedidos {
3      public static void main(String[] args) {
4          Pedido pedido = new Pedido();
5
6          pedido.inicializarImposto(); // adição de left
7          System.out.println("Divisor");
8          pedido.valorImposto = 0.20; // alteração de right
9
10         System.out.println("Valor do imposto: " + pedido.valorImposto);
11     }
12 }
```

Pedido.java

```java
1  public class Pedido {
2      public double valorImposto;
3
4      public void inicializarImposto() {
5          this.valorImposto = 0.15; // altreração de left
6      }
7  }
8
```

In line 6 of ‘ProcessadorDePedidos.java’, left adds a method call, which in ‘Pedido.java’ modifies the variable ‘valorImposto’ in line 5, modified again by right in line 8 of ‘ProcessadorDePedidos.java’

C2 - In one file, `left` adds a method call to another file, where a variable is defined/modified by `left`. In that file, `right` modifies that variable again.

ProcessadorDePedidos.java

```java
1  // Arquivo: ProcessadorDePedidos.java
2  public class ProcessadorDePedidos {
3      public static void main(String[] args) {
4          AjustadorDePreco ajustador = new AjustadorDePreco();
5          ajustador.processarPedido(); // Adição de left
6          System.out.println("Preço final: " + AjustadorDePreco.preco);
7      }
8  }
```

AjustadorDePreco.java

```java
1  // Arquivo: AjustadorDePreco.java
2  public class AjustadorDePreco {  no usages
3      public static double preco = 0.0;  2 usages
4
5      public void processarPedido() {  no usages
6          preco = 100.0 * 1.05; // Alteração de left
7          System.out.println("Divider");
8          preco = 120.0; // Alteração de rigth
9      }
10 }
```

In line 5 of the file ‘ProcessadorDePedidos.java’, left calls the method ‘processarPedido()’ in the file ‘AjustadordePreco.java’. In line 7 of ‘AjustadordePreco.java’, left modifies the variable ‘preco’, which is modified again by right in line 8.

D2- In file 1 there is a method call to file 2, where left defines/changes a variable. Additionally, there is another method call in file 1 to file 2, where right changes the same variable that left defined/changed.

Processador.java

```java
1  // Arquivo: Processador.java
2  public class Processador {
3      public static void main(String[] args) {
4          Ajustador ajustador = new Ajustador();
5
6          ajustador.aplicarTaxa(); // adição de left
7
8          ajustador.aplicarDescontoFixo(); // adição de right
9
10         System.out.println("Preço final atual: " + Ajustador.precoFinal);
11     }
12 }
```

Ajustador.java

```java
1  // Arquivo: Ajustador.java
2  public class Ajustador {
3      public static double precoFinal = 100.0;
4
5      public void aplicarTaxa() {
6          precoFinal = precoFinal * 1.10; // alteração de left
7      }
8
9      public void aplicarDescontoFixo() {
10         precoFinal = 90.0; // alteração de right
11     }
12 }
```

On line 6, in ‘Processador.java’ there is a method call to ‘Ajustador.java’, where left modifies the variable ‘preçoFinal’ on line 6. On line 8 in ‘Processador.java’ there is a call to the method ‘apliciarDescontoFixo()’ in ‘Ajustador.java’, where right modifies the same variable ‘preçoFinal’ on line 10.

E2- In file 1, left adds a method call to file 2, where a variable is defined. Still in file 2, right makes a method call back to file 1, where it modifies the same variable defined by left.

```
1   public class Main {  no usages  ykarocin *
2
3       public static void updateCounter() {  no usages  new *
4           State.counter = 20; // Right change
5       }
6
7       public static void main(String[] args) {  no usages  ykarocin *
8           Worker.initialize(); // Left change
9       }
10  }
11
```

```
1   public class Worker {  no usages  ykarocin *
2
3       public static void initialize() {  no usages  new *
4           State.counter = 10;  // left change
5           System.out.println("Counter Value: " + State.counter);
6           Main.updateCounter(); // right change
7   }
```

On line 8 of “Main.java”, the left method calls “Worker.java” and changes the value of counter. On line 6 of “Worker.java”, the right method calls “Main.java” and changes the value of counter again on line 4.

A3- In file 1, there is a method call to file 2, where left defines/modifies a variable. There is also a file 3, where right modifies the same variable defined/modified by right.

Faturador.java

```java
1 // Arquivo: Faturador.java
2 public class Faturador {
3     public static void main(String[] args) {
4         CalculadoraDeItens calc = new CalculadoraDeItens();
5         calc.calcularItens(); // Adição de left
6         System.out.println("Valor da fatura: " + CalculadoraDeItens.valorFatura);
7     }
8 }
```

CalculadoraDeItens.java

```java
1 // Arquivo: CalculadoraDeItens.java
2 public class CalculadoraDeItens {
3     public static double valorFatura = 0.0;
4 
5     public void calcularItens() {
6         valorFatura = 100.0; // alteração de left
7     }
8 }
```

AjustadorDeTaxas.java

```java
1 // Arquivo: AjustadorDeTaxas.java
2 public class AjustadorDeTaxas {
3     static {
4         CalculadoraDeItens.valorFatura = 120.0; // Alteração de right
5     }
6 }
```

In line 5 of ‘Faturador.java’ there is a method call to the file ‘CalculadoraDeItens.java’, where the variable ‘valorFatura’ is modified by left in line 6. In ‘AjustadorDeTaxas.java’ in line 4, right modifies the same variable ‘valorFatura’.

B3- In file 1 there is a method call to file 2, where `left` modifies/defines a variable. In file 3 there is also a method call to file 2, where `right` modifies the variable modified by `left` again.

Contador.java

```java
// Arquivo: Contador.java
public class Contador {
    public static void main(String[] args) {
        AjusteDeValores ajuste = new AjusteDeValores();
        ajuste.definirSaldoPadrao(); // Alteração de left
        System.out.println("Saldo inicial: " + AjusteDeValores.saldo);
    }
}
```

AjusteDeValores.java

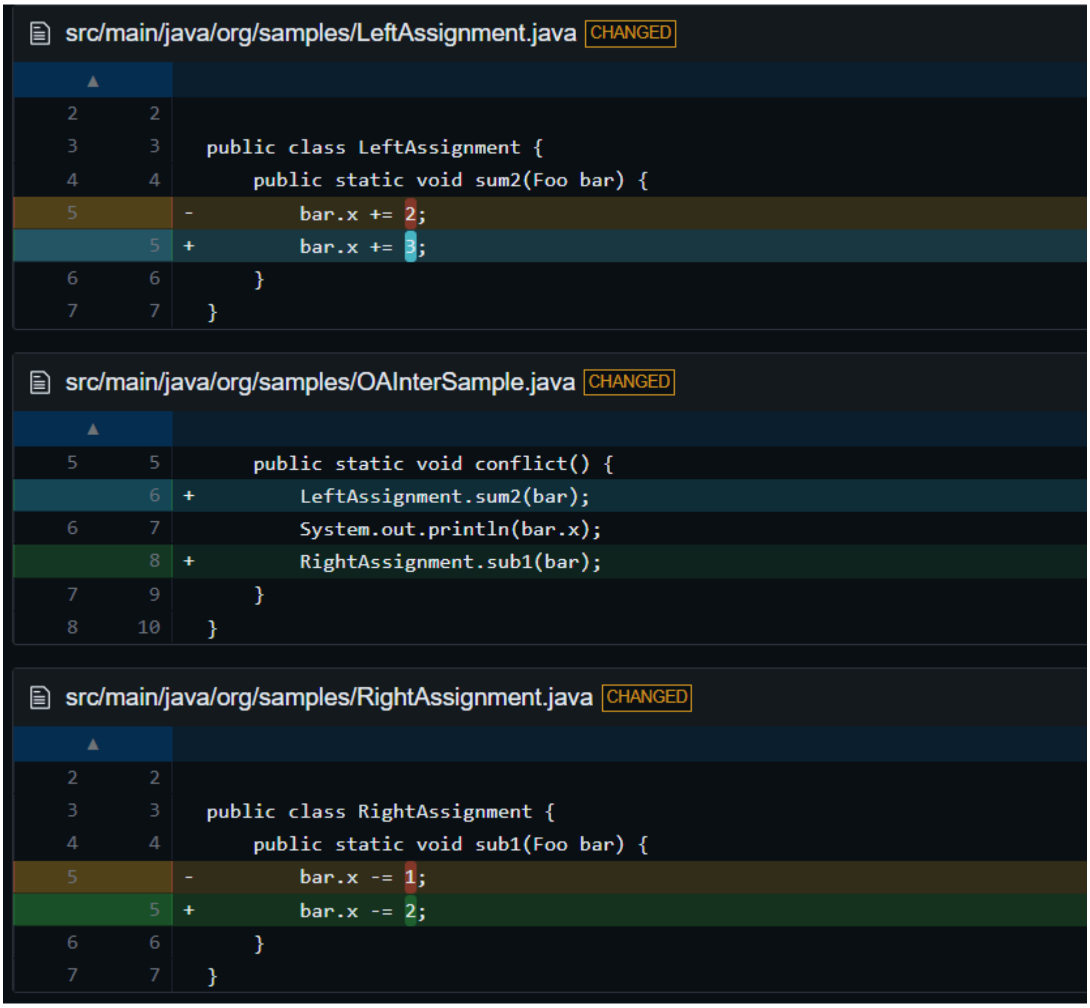
```java
// Arquivo: AjusteDeValores.java
public class AjusteDeValores {
    public static double saldo = 0.0;

    public void definirSaldoPadrao() {
        saldo = 1000.0; // Alteração de left
    }

    public void aplicarBonusDeFidelidade() {
        saldo = 1500.0; // Alteração de right
    }
}
```

Relatorio.java

```java
// Arquivo: Relatorio.java
public class Relatorio {
    public void gerarRelatorioMensal() {
        AjusteDeValores ajuste = new AjusteDeValores();
        ajuste.aplicarBonusDeFidelidade(); // Alteração de right
    }
}
```

In the file ‘Contador.java’ there is a method call on line 5 to the file ‘AjusteDeValores.java’ where left modifies the variable ‘saldo’ on line 6. In ‘Relatorio.java’ there is a method call on line 4 to ‘AjusteDeValores.java’, where on line 10, right modifies the variable ‘saldo’ previously modified by left again.

C3 - In the central file there are two method calls, one to the upper file, where left changes the value of a variable, and another to the bottom file, where right changes the same variable.



src/main/java/org/samples/LeftAssignment.java CHANGED

```
2   2
3   3      public class LeftAssignment {
4   4          public static void sum2(Foo bar) {
5       -          bar.x += 2;
    5   +          bar.x += 3;
6   6          }
7   7      }
```

src/main/java/org/samples/OAInterSample.java CHANGED

```
5   5          public static void conflict() {
    6   +          LeftAssignment.sum2(bar);
6   7              System.out.println(bar.x);
    8   +          RightAssignment.sub1(bar);
7   9          }
8   10     }
```

src/main/java/org/samples/RightAssignment.java CHANGED

```
2   2
3   3      public class RightAssignment {
4   4          public static void sub1(Foo bar) {
5       -          bar.x -= 1;
    5   +          bar.x -= 2;
6   6          }
7   7      }
```

In line 6 of the file ‘OaInterSample.java’, left makes a method call to a method in the file ‘LeftAssignment.java’, where it changes the value of the variable ‘bar.x’ in line 5. Then, in line 8 of ‘OAInterSample.java’, right makes a method call to a method in the file ‘RightAssignment.java’, where it also changes the value of the variable ‘bar.x’ in line 5, generating an OA interference from line 5 of ‘RightAssignment.java’ to line 5 of ‘LeftAssignment.java’.

D3- In file 1, left adds a method call to file 2 where it changes/sets the value of a variable. Later in file 2, right adds a method call to file 3, where it changes the same variable that left changed/set..

```
1  public class Main {  no usages  ykarocin *
2      public static void main(String[] args) {
3          Worker.initialize(); // Left change
4      }
5  }
```

```
1  public class Worker {  no usages  ykarocin *
2      public static void initialize() {  no usages  new *
3          State.counter = 10;  // left change
4          System.out.println("Counter Value: " + State.counter);
5          State.updateCounter(); // right change
6  }
```

```
1  public class State {  1 usage  new *
2      public static int counter = 0;  1 usage
3
4      public static void updateCounter() {  no usages
5          State.counter = 20; // Right change
6      }
7  }
```

In line 3 of “Main.java”, left adds a method call to “Worker.java” that changes the value of “counter” in line 3. In line 5 of “Worker.java”, right adds a method call to “State.java” that changes the value of “counter” in line 6.

A4- In file 1 there is a method call to file 2, where left modifies/defines a variable.In file 3 there is a method call to file 4, where `right` modifies the variable defined/modified by `left`.

GerenciadorDeProducao.java ×

```
public class GerenciadorDeProducao {  no usages
    public static void main(String[] args) {
        CalculadoraDeCusto calc = new CalculadoraDeCusto();
        calc.calcular(); // Adição de left

        OtimizadorDeProcesso otimizador = new OtimizadorDeProcesso();
        otimizador.otimizar();

        System.out.println("Custo total: " + CalculadoraDeCusto.custoTotal);
    }
```

CalculadoraDeCusto.java

```java
1 public class CalculadoraDeCusto {
2     public static double custoTotal = 0.0;
3 
4     public void calcular() {
5         custoTotal = 500.0; // Alteração de left
6     }
7 }
```

OtimizadorDeProcesso.java

```java
1 public class OtimizadorDeProcesso {
2     public void otimizar() {
3         RelatorioDeEficiencia relatorio = new RelatorioDeEficiencia();
4         relatorio.gerar(); // Adição de right
5     }
6 }
```

RelatorioDeEficiencia.java

```java
1 public class RelatorioDeEficiencia {
2     public void gerar() {
3         CalculadoraDeCusto.custoTotal = 450.0; // Alteração de right
4     }
5 }
```

In line 4 of ‘GerenciadorDeProducao.java’, there is a method call to ‘CalculadoraDeCusto.java’, where on line 5, left modifies the variable ‘custoTotal’. In ‘OtimizadorDeProcesso.java’, on line 4, there is a method call to ‘RelatorioDeEficiencia.java’, where right modifies the variable ‘custoTotal’ again on line 3.